TxBench-PP: Verifiable Evaluation of AI Agents on Preclinical Pharmacology

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ABSTRACT

We introduce TxBench-PP, a verifiable benchmark for evaluating AI agents on small-molecule preclinical pharmacology. TxBench-PP is the first focused slice of a broader TherapeuticsBench effort organized across drug-discovery stages and therapeutic modalities. It targets recurring cluster-style decisions: whether agents can recover decision-relevant conclusions from supplied assay evidence rather than recalling drug labels, ranking by a single metric, or applying generic checklists. The current task inventory maps 70 evaluations to a nine-stage Assays Roadmap. Seven stages have coverage, with strongest current coverage in causal target validation and perturbational mechanism-of-action / pharmacodynamic reasoning, single-source coverage in screening, pharmacogenomics, safety, and translational efficacy, and a near-empty compound-target characterization stage. Agents receive realistic workflow snapshots, inspect files in a coding environment, and return structured answers graded deterministically. **[TODO:** Replace this final paragraph with model-run results: number of evaluations, model-harness pairs, trajectories, top pass rates, and the main failure pattern.]

TxBench-PP in TherapeuticsBench

TxBench-PP is the current small-molecule preclinical pharmacology slice of a broader TherapeuticsBench roadmap. The full benchmark family is organized by drug-discovery stage and therapeutic modality; this paper focuses on repeated, verifiable cluster decisions in small-molecule programs. These decisions sit between initial discovery and clinical development: whether a molecule engages the intended biology, whether the evidence supports its mechanism, whether exposure can cover the relevant potency range, and whether safety signals block advancement.

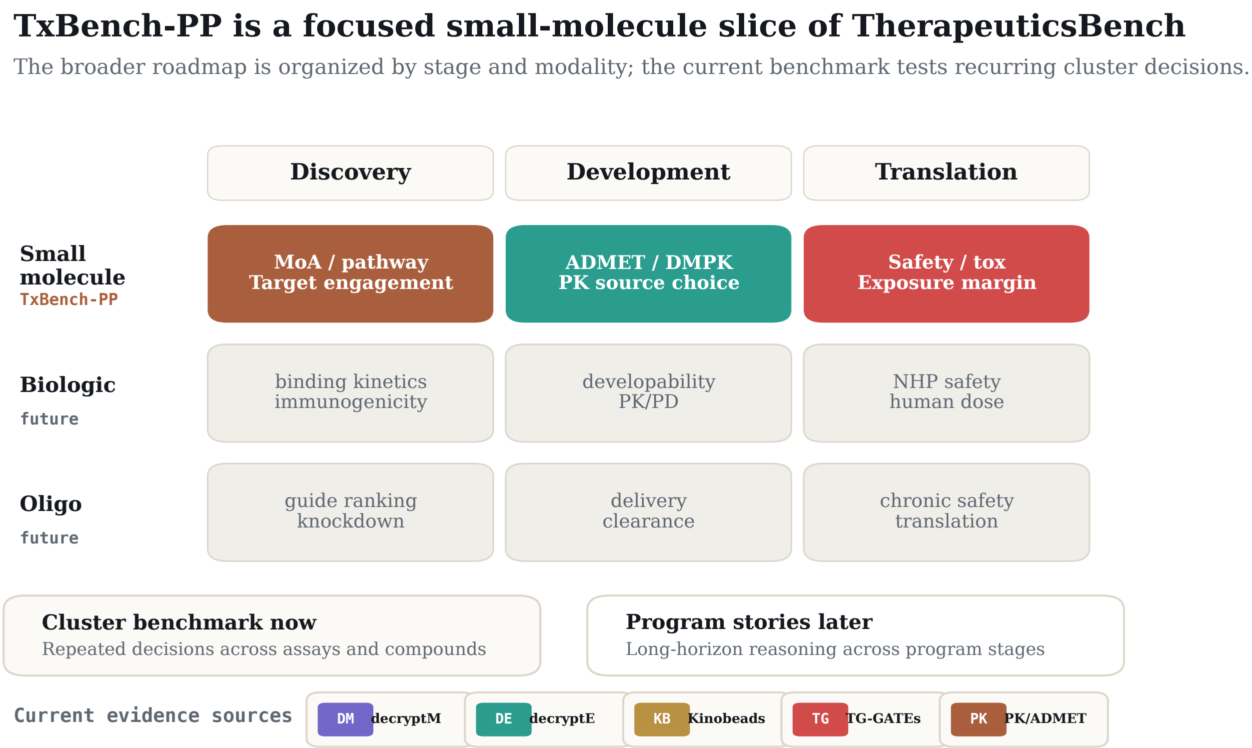

Figure 1: TxBench-PP as a TherapeuticsBench slice. TherapeuticsBench is intended to span discovery, development, and translation across small molecules, biologics, and oligonucleotides. TxBench-PP covers the small-molecule preclinical pharmacology slice as a cluster benchmark: repeated local decisions across assay families and compounds. Program-story benchmarks that follow one therapeutic program across discovery, development, and translation are reserved for future work.

Introduction

In preclinical drug discovery, experiments are slow and expensive, but many program-limiting decisions are interpretive: which signal is real, which compound should advance, and which liability should block a program. Preclinical pharmacology converts assay evidence into these decisions. A compound can have an attractive potency value and still be a poor candidate because the apparent mechanism is an artifact, the engaged protein is a complex contaminant, the safety signal is protocol-specific, or total exposure does not translate into unbound target coverage.

AI agents can inspect files, write analysis scripts, and invoke scientific software, but their conclusions often remain vulnerable to plausible shortcuts. In drug discovery, those shortcuts have familiar forms: importing the known mechanism of a blinded compound, trusting the largest apparent effect without quality control, ranking by potency or total exposure alone, or applying a textbook safety threshold outside the protocol where it was defined.

We introduce TxBench-PP, a benchmark for small-molecule preclinical pharmacology decisions. It should be read as a focused cluster benchmark within the broader TherapeuticsBench roadmap rather than as a complete test of drug discovery. Each evaluation supplies the agent with realistic analysis artifacts from one stage of the preclinical pharmacology evidence chain and asks for a structured, deterministically gradable conclusion. The benchmark tests whether an agent can recover the result supported by the provided evidence, not whether it can recall a drug class, run a default workflow, or optimize for a single number.

The current benchmark maps 70 evaluations to a nine-stage Assays Roadmap. Coverage is strongest in causal target validation and perturbational mechanism-of-action / pharmacodynamic reasoning; the clearest build gap is compound-target characterization. The underlying sources still matter: several covered stages are dominated by a small number of datasets, so diversifying assay sources is a central part of the benchmark roadmap. **[TODO:** Add final run summary once result aggregation is complete.]

Benchmark Construction

To construct TxBench-PP, we decompose preclinical pharmacology workflows into short, gradeable scientific decisions. Examples include identifying a pathway-coherent mechanism from dose-resolved phosphoproteomics, separating true chemoproteomic target engagement from complex contaminants, reconciling conflicting clearance measurements, and integrating blood chemistry, pathology, and survival into an in vivo safety call.

This makes TxBench-PP a cluster-style benchmark: the same kind of decision is tested repeatedly across assay contexts, compounds, and evidence sources. Future TherapeuticsBench story evaluations can test the complementary skill of following one therapeutic program across discovery, development, and translation. TxBench-PP instead asks whether agents can make the local scientific decisions that those longer program stories depend on.

Each task includes a snapshot of assay outputs immediately before a target decision, metadata needed to interpret the files, a high-level task description, and a deterministic grader. Tasks specify what result should be recovered rather than prescribing the exact analysis method. Procedural details are included only when they are part of the scientific decision, such as whether free or total exposure should be compared to a pharmacologic threshold.

The benchmark follows three design criteria. First, tasks must be verifiable: answers are checked by structured label matching, numerical tolerances, rank-order checks, or all-of field comparisons. Second, tasks should be decision-relevant: the answer should correspond to a real go/no-go or prioritization judgment in preclinical pharmacology. Third, tasks should resist shortcuts: agents should need to inspect the supplied evidence, account for quality issues, and avoid relying on memorized drug labels or generic checklists.

We separately track decision-blocking gaps: issues that should change the program decision if recognized, such as a tox-species mismatch, insufficient free-drug coverage, a hook effect, a contaminated chemoproteomic hit, or survivor bias in pathology sampling. These labels are intended to support diagnostic analysis after endpoint grading.

Evaluation Inventory

The current suite is organized around the Assays Roadmap rather than around source datasets. Each evaluation is assigned to the roadmap stage that best matches its primary biological skill. The current inventory includes 70 evaluations across seven of the nine stages. S1 disease/model characterization and S4 human genetic target support are deferred for this release; S4 is better handled by GenomicsBench.

The roadmap view changes the benchmark story. Coverage is broad by stage count, but narrow by dataset diversity. Causal target validation and perturbational MoA/PD are ready for v1. Screening, pharmacogenomics, safety, and translational efficacy have useful coverage but are single-source dominated. The largest in-scope gap is S7 compound-target characterization: binding, potency, selectivity, and cellular target engagement are almost absent.

Evaluation inventory follows the nine-stage Assays Roadmap

Coverage is broad by stage count, but several covered stages remain single-source dominated.

70 evals | **7/9** stages covered | **S7** build gap

Stage		Evals	Status	Roadmap interpretation
S1	Disease / model characterization	0	Deferred	out of v1 scope
S2	Primary screening & hit confirmation	7	Diversify	PRISM-heavy
S3	Drug response / pharmacogenomics	11	Diversify	mostly PRISM
S4	Human genetic target support	0	Deferred	GenomicsBench
S5	Causal target validation	12	Ready	v1 coverage
S6	Perturbational MoA / PD	17	Ready	strongest coverage
S7	Compound-target characterization	1	Build	largest gap
S8	Developability & safety	16	Diversify	cardiotox-heavy
S9	Translational efficacy	6	Diversify	one NSCLC source

Counts and statuses are from the Assays Roadmap coverage mapping. S1 and S4 are intentionally deferred for this release.

Figure 2: Evaluation inventory by Assays Roadmap stage. Current TxBench-PP evaluations are mapped to the roadmap stage that best captures the biological skill being tested. The inventory is broad across stages but uneven across assay families: S5 and S6 are ready for v1, S7 is the clearest build gap, and S2, S3, S8, and S9 need source diversification.

Table 1: Roadmap-stage inventory. Counts are from the Assays Roadmap coverage mapping of 70 current evaluations. "Ready" stages are sufficient for v1; "Diversify" stages have useful but source-narrow coverage; "Build" marks the largest missing pillar.

Roadmap stage	Evals	Status	Assay emphasis and next gap
S1. Disease / model characterization	0	Deferred	Model subtype, expression, mutation/CNV context, pathway state, IHC/IF, RNAscope, and model QC are out of the current build queue.
S2. Primary screening / hit confirmation	7	Diversify	PRISM-heavy. Add biochemical or reporter HTS, Z-prime and signal-window plate QC, counterscreens, and frequent-hitter filtering.
S3. Drug response / pharmacogenomics	11	Diversify	Mostly PRISM. Add GDSC, CTRP, CCLE, Beat AML, organoid screens, single-cell perturbation, and censored-curve handling.
S4. Human genetic target support	0	Deferred	GWAS, QTL, colocalization, Mendelian randomization, and tissue-of-action matching are assigned to GenomicsBench.
S5. Causal target validation	12	Ready	Strong v1 coverage. Optional expansion: Perturb-CITE-seq, rescue assays, variant-function classification, and FACS sort-bin screens.
S6. Perturbational MoA / PD	17	Ready	Strong v1 coverage. Optional expansion: LINCS/CMap, DRUG-seq, Seahorse, cytokine panels, reporter dose-response, and washout PD.
S7. Compound-target characterization	1	Build	Highest-priority gap. Add ChEMBL/BindingDB potency curation, selectivity panels, CETSA/NanoBRET, Kinobeads, and biochemical-to-cellular potency comparison.
S8. Developability / safety	16	Diversify	Cardiotox dominated. Add measured hERG, DILI, Ames/micronucleus, permeability, metabolic stability, CYP/DDI, Tox21/ToxCast, and therapeutic-index reasoning.
S9. Translational efficacy	6	Diversify	Single NSCLC source. Add organoid/patient concordance, PDX pharmacology, immune co-culture, ctDNA, responder-fraction, and exposure-response PK/PD.

Cross-stage gap. The roadmap also identifies a missing class of end-to-end go/no-go evaluations: tasks that require an agent to weigh efficacy, selectivity, target engagement, exposure, and safety together. The current inventory covers many of these component decisions, but future TxBench-PP tasks should test multi-stage synthesis across at least three evidence blocks.

Results

No agent reliably recovers preclinical pharmacology decisions

[**TODO:** Topline paragraph from the model leaderboard (`fig1_leaderboard`): state the scope (11 model families, 100 evaluations $\times$ 3 rounds on the Pi harness), the pass-rate definition, and the ordering from the leader (claude-opus-4-8, $\sim$60%) down to the floor (grok-4.3, $\sim$19%), with raw counts and confidence intervals. Emphasize that even the best system leaves roughly 40% of decisions unrecovered.]

Most failures map to a compact failure-mode taxonomy

To understand why agents fail, we read every failing (model, evaluation) run on the Pi harness and labeled it against the evaluation's documented scientific intent, resulting in 1,834 labeled failures. Each failure is classified along two complementary axes. The *scientific error* captures what is wrong with the answer, and the *behavioral cause* captures how the model arrived there (Figure 3).

The scientific axis comprises five recurring error families (Figure 3A, restricted to the $n = 1{,}436$ failures that contain a scientific error). The largest is method, which resolves into two main sub-modes plus a small tail. A *skipped-QC* method error (17%) omits a required preprocessing or quality-control step. In one case, a model aggregated CRISPR guides to gene-level scores without first dropping guides that map to multiple loci, so paralog-family artifacts flooded the essential-gene list. A *wrong-reference-frame* method error (21%) runs the pipeline but compares against the wrong baseline, comparator, or normalization. A small *wrong-assay-layer* tail (2%) reads the wrong assay feature or modality entirely. A *calibration* error (31%) draws a cutoff a domain expert would set differently. In one competition-binding panel, models carried forward all 33 nonzero kinases when only the 8 strong binders above a clear affinity gap were supported. A *perception* error (10%) misreads a raw image, for example counting apoptotic debris as viable organoids or mis-segmenting two-channel colocalization. *Prior over data* (10%) answers from textbook or famous biology against the staged evidence. One model computed the strongest data-supported biomarker, a vanadium compound correlated with SLC26A2 at $r = -0.48$, and then discarded it for the textbook elesclomol-FDX1 cuproptosis pair. *Over-reading* (10%) states a stronger claim than the data supports, for example reporting tumor regression from a sub-baseline median when the per-animal measurements show only growth arrest. Method and calibration together account for 71% of scientific errors.

The behavioral axis records how the failure happened, over all $n = 1{,}834$ runs (Figure 3B). Two-thirds (67%) are *engaged, analytically wrong*. The model did the work in good faith and made an analytical error. The remaining third splits into one evidence-handling failure and three process failures. *Prior-driven override* (6%) is the one behavior in which the model engaged the data, often computed the supporting result, and then deferred to a textbook or famous-biology prior. It is the behavioral signature of the scientific *prior over data* error, and the two coincide almost exactly. *Weak commitment* (9%) performs the right analysis but submits the wrong final answer. In one run a model's reasoning correctly identified CTNNB1 as the supported dependency while its output named E2F3. *Abandoned-correct* (12%) is the sharpest case. The model demonstrably had a passing result within reach and then committed to a worse one. *Under-investment* (6%) reaches an answer with little tool use or verification, for example a mitotic-kinase family call written from prior knowledge after merely listing the compounds. Per-run effort separates these modes. Under-investment is the only one written with markedly less work (median 6 reasoning steps against 14–17 for the others). Abandoned-correct and prior-driven override occur in the *most* effortful runs (median 16 steps), so they are not relieved by additional compute. That a third of failures are non-analytical, and that abandoned-correct alone covers 212 runs in which the right result was within reach, implies that a meaningful fraction is recoverable through better verification and final-answer selection rather than added capability.

The failures have a unifying frame. These are fluency-optimized systems applied to a domain that rewards adversarial restraint. The fluent move is confident, coherent, and consistent with prior knowledge, while the right experimental move defers to the data over the textbook, draws conservative cuts, and distrusts its own first answer. TxBench-PP's traps sit where these diverge, so the failures are a systematic readout of that misalignment. The models have the knowledge and the tooling, and often compute the right intermediate result, but they lack the epistemic stance of an experimentalist, namely prior-skepticism, calibrated commitment, and self-verification.

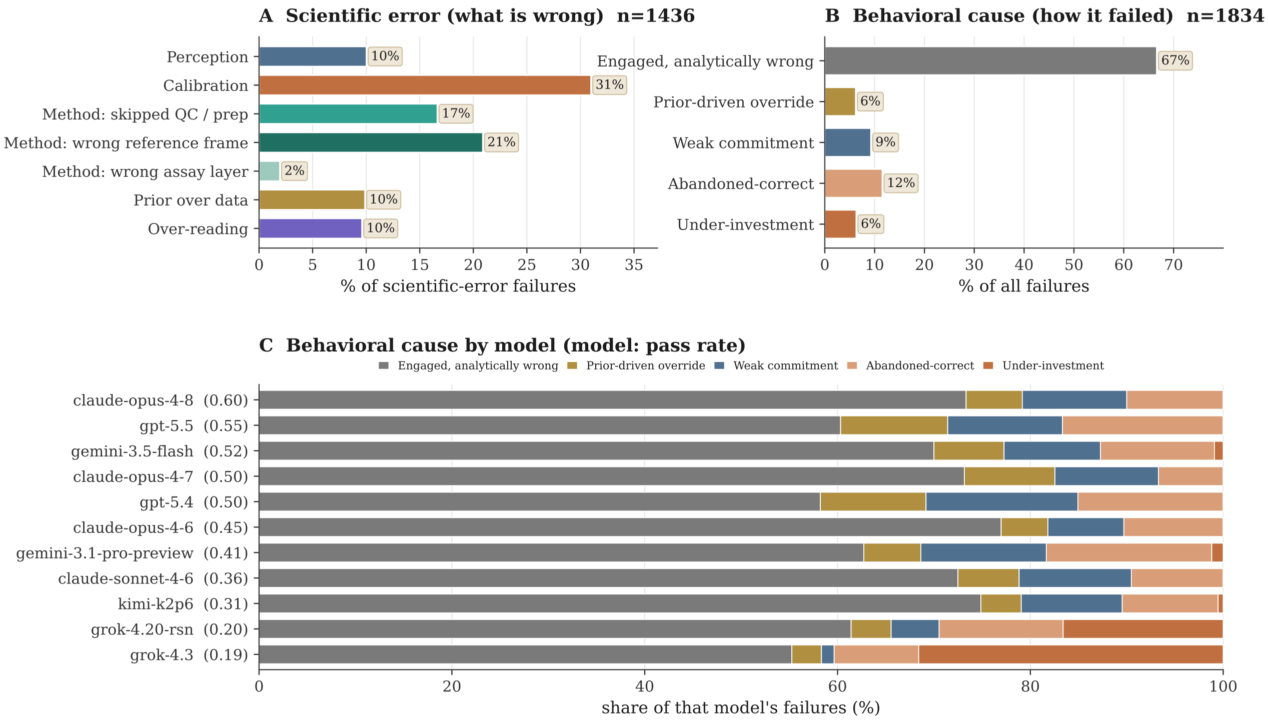

Figure 3: **A two-axis failure taxonomy.** All 1,834 failing (model, evaluation) runs on the Pi harness, labeled against each evaluation's scientific intent and classified along two complementary axes. (**A**) The scientific-error axis (what is wrong with the answer) is dominated by method and calibration errors (71% combined), which are general analytical skills rather than domain knowledge; the method family resolves into a skipped-QC/preprocessing sub-mode and a larger wrong-reference-frame sub-mode. (**B**) The behavioral-cause axis (how the model failed) shows that a third of failures are non-analytical: a prior-driven override of the model's own evidence, plus three process breakdowns (weak commitment, abandoned-correct, under-investment). (**C**) The behavioral mix is a model signature: under-investment dominates only the weakest model, while the strongest models fail almost entirely through engaged, analytically wrong reasoning. A scientific-error $\times$ behavioral-cause cross-tabulation is provided in the supplement (`fig_fm_crosstab`).

The dominant failure mode shifts characteristically across pipeline stages

[TODO: From the therapeutic-pipeline-stage breakdown (`fig3_by_tx_stage` and `_stage_failure_maps.md`): report per-stage pass rates and the mode that dominates each stage (e.g. S5 causal target validation is skipped-QC heavy; S9 translational efficacy is threshold-over-inclusion heavy; S2 screening shows convergent quantitative misreads). Make clear the failure signature is stage-specific rather than uniform.]

Some task categories remain far harder than others

[TODO: From the task-type breakdown (`fig2_by_task_type`, with `fig2b`/`fig2c` for the per-model view and heatmap): pass rate ranges from exposure-PK (∼71%) down to cheminformatics (∼17%); include the worst-to-best per-model whiskers to show the spread within each category.]

The strongest models win through disciplined scientific process

[TODO: From the per-model signatures (`_per_model_signatures.md`) and the cost/steps frontier (`fig5_cost_frontier`): spending does not buy accuracy (the Grok models under-invest; sonnet over-spends without yield). What separates strong runs is process discipline (reading images then verifying in code, framing the question correctly, drawing a defensible threshold, and committing to a discrete answer), not domain knowledge, which every model already has.]

Failures translate directly into wrong go/no-go decisions

[TODO: Use the downstream-consequence lists in `_stage_failure_maps.md` to tie representative failures to concrete program errors (e.g. advancing a non-selective compound as selective, a wrong cytostatic-vs-cytotoxic call, a missed safety liability, a contaminated target-engagement hit carried forward). Ground each in an exact eval name once the public manifest is frozen.]

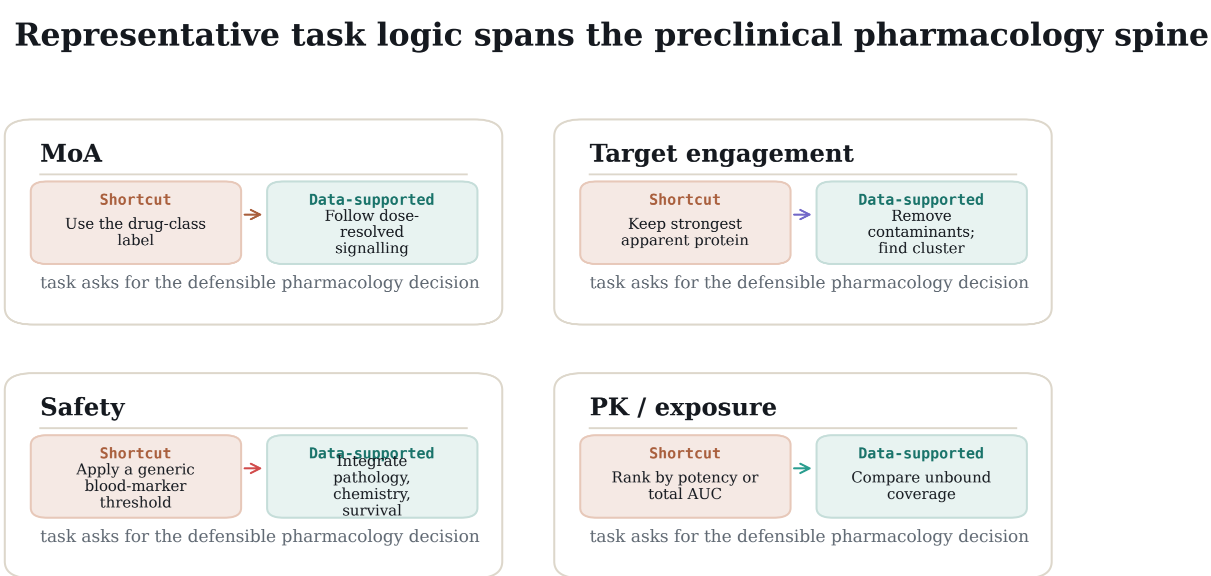

Figure 4: Representative task logic. Across mechanism, target engagement, safety, and exposure decisions, the correct answer is the one supported by the supplied assay evidence, not the familiar shortcut.

The agent harness has a surprising impact on model performance

[**TODO:** From the harness-effect comparison (`fig4_harness_effect`): running the same model under different scaffolds (Pi vs Codex/CC) shifts the task-matched pass rate by roughly 3–12 points, with some models far more harness-sensitive than others (e.g. claude-opus-4-7) even though the overall ordering is largely preserved. Note the implication for fair model comparison: reported scores are a property of the model-plus-harness, not the model alone.]

Discussion

TxBench-PP measures a narrow but important part of drug discovery: whether an agent can turn preclinical pharmacology evidence into a defensible local decision. This scope is intentionally smaller than end-to-end therapeutic discovery. It is a small-molecule cluster benchmark within the broader TherapeuticsBench roadmap, not a claim to cover every stage or modality. It does not yet test target genetics, full disease-model selection, primary screening operations, broad clinical translation, or non-small-molecule modalities. The narrower scope makes the benchmark verifiable while still covering decisions that directly affect compound advancement.

The central hypothesis is that current agents will often operate the tools of preclinical analysis but fail at the scientific judgment layer. If a model selects a mechanism from a drug label, trusts a contaminated target-engagement signal, ignores survivor bias, or ranks by total exposure rather than unbound coverage, it may produce a fluent and computationally elaborate answer that is not the pharmacology decision supported by the data.

[**TODO:** After results are available, add a paragraph comparing endpoint pass rates, partial-credit or field-level scores if available, and trajectory review. Emphasize what capability is missing and which tasks show meaningful progress.]

Future versions can extend the benchmark upstream to target rationale and primary screening, sideways to pharmacogenomics and disease-model characterization, and across modalities to antibodies, ADCs, degraders, oligonucleotides, cell therapies, and gene therapies. A complementary long-horizon TherapeuticsBench track can follow program stories across discovery, development, and translation. For the current paper, the clean claim should remain: TxBench-PP is a verifiable cluster benchmark for small-molecule preclinical pharmacology decisions.

Methods

Task construction

Candidate evaluations are derived from realistic preclinical pharmacology workflow states. A task is retained when the target conclusion can be recovered from the supplied files, expressed through a constrained final answer, and graded deterministically. Tasks should be revised or removed when the prompt over-specifies the method, when multiple reasonable analyses produce incompatible decisions, or when the grader cannot distinguish the supported pharmacology conclusion from a plausible but unsupported answer.

Agent runs and aggregation

[**TODO:** Insert model-harness list, number of attempts per evaluation, execution environment, parsing rules, and how invalid or missing outputs are counted.]

Statistical analysis

[**TODO:** Insert primary metric and confidence interval convention. If following SpatialBench/EpiBench, report evaluation-weighted endpoint pass rate, raw pass counts for interpretability, and Student t confidence intervals over evaluation-level replicate means.]

Data availability

[**TODO:** Insert public repository, public example evaluations, result summaries, and trajectory availability language.]

References

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